

GRAND MARAIS HARBOR, MN, USA

WMO: 997259

Lat: 47.7483N

Lon: 90.3415W

Elev: 186

StdP: 99.11

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -24.4	(c) -21.7	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 7.7	(c) 24.7	(d) N/A	(e) 22.8	(f) N/A	(g) 21.4	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) N/A	(b) N/A	(c) N/A	(d) DB	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A	(q) N/A
(a) N/A	(b) N/A	(c) N/A	(d) WB	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A	(q) N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	4.3	-9.6	-9.0	-2.4	3.2	8.3	12.2	16.6	16.8	13.5	7.5	0.2	-6.5	(6)
(7)		DBStd	10.31	7.08	5.77	5.62	3.82	2.82	2.82	3.07	2.44	3.30	3.66	5.30	5.85	(7)
(8)		HDD10.0	2718	606	532	384	204	68	9	0	0	8	97	296	513	(8)
(9)		HDD18.3	5156	864	766	643	453	312	186	71	60	147	337	545	771	(9)
(10)		CDD10.0	640	0	0	1	1	14	75	206	212	113	18	0	0	(10)
(11)		CDD18.3	36	0	0	0	0	0	2	18	13	3	0	0	0	(11)
(12)		CDH23.3	122	0	0	0	0	1	14	63	39	5	0	0	0	(12)
(13)		CDH26.7	15	0	0	0	0	0	3	6	6	1	0	0	0	(13)
(14)	Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(14)
(15)	Precipitation	PrecAvg	685	31	21	31	53	68	91	82	70	77	84	53	36	(15)
(16)		PrecMax	912	63	49	84	194	121	213	204	142	163	228	98	76	(16)
(17)		PrecMin	528	8	5	10	10	22	10	32	22	18	33	15	9	(17)
(18)		PrecStd	108	15	12	17	38	25	44	42	29	35	47	25	18	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.8	3.6	11.7	15.8	19.5	25.5	27.5	27.0	23.5	18.7	12.0	5.0	(19)
(20)		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(20)
(21)		2%	DB	2.8	1.8	8.4	12.5	16.6	20.9	25.3	24.1	21.4	15.7	10.1	3.6	(21)
(22)		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(22)
(23)		5%	DB	1.6	0.5	6.3	10.1	14.5	18.8	23.3	22.2	19.7	14.2	8.6	2.8	(23)
(24)		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(24)
(25)		10%	DB	0.3	-0.8	4.4	8.2	12.7	16.9	21.7	20.8	18.4	12.8	7.4	1.7	(25)
(26)		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(27)
(28)		MADB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(28)
(29)		2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(29)
(30)		MADB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(30)
(31)		5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(31)
(32)		MADB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(32)
(33)		10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(33)
(34)		MADB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(34)
(35)	Mean Daily Temperature Range	MDBR	8.0	8.0	7.2	6.9	7.7	8.3	8.4	7.7	6.7	5.9	6.0	6.9	(35)	
(36)		5% DB	MCDBR	6.7	6.8	8.6	9.6	10.5	11.3	10.5	9.7	8.4	7.8	6.3	5.1	(36)
(37)		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(37)
(38)		5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(38)
(39)		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.260	0.269	0.301	0.345	0.364	0.386	0.396	0.381	0.365	0.318	0.292	0.266	(40)	
(41)		taud	2.431	2.398	2.399	2.381	2.415	2.378	2.398	2.453	2.468	2.562	2.534	2.459	(41)	
(42)		Ebn at Noon	840	907	924	908	899	877	862	862	844	841	796	786	(42)	
(43)		Edh at Noon	64	83	98	111	112	117	113	103	92	72	60	56	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.20	2.19	3.49	4.69	5.40	5.81	6.05	5.13	3.76	2.25	1.21	0.91	(44)	
(45)		RadStd	0.15	0.24	0.32	0.49	0.36	0.42	0.27	0.35	0.24	0.26	0.15	0.09	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (27 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page